

Braden Addison Kennedy

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EDUCATION

Cornell University

M.Eng. Mechanical Engineering

Ithaca, NY

Expected Dec. 2027

Cornell University

B.S. Engineering Physics, Minor in Mechanical Engineering

Ithaca, NY

Expected May 2027

Cumulative GPA: 3.98

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern

Summer 2025, Winter 2026, Summer 2026

Acbotics Research

Falmouth, MA

- Led mechanical design of the AcSample eDNA sampler through 3 iterations (two prototypes, one full-production model), redesigning housings, tubing, and sealing for improved modularity, manufacturability, and reliability across a range of customer configurations
- Fabricated and tested 15+ AcSample units (including shallow and oil-filled variants) and other custom underwater speakers and sensors, integrating electronics and validating seals, wiring, software behavior, and structural performance, and writing user and production manuals
- Deployed and commissioned 12-channel hydrophone arrays and underwater speaker systems for research groups, training customers on data collection procedures for their published work

Undergraduate Researcher, Carl Franck Lab

Dec 2024 - Present

Cornell University

Ithaca, NY

- Engineered and characterized an experimental apparatus to measure intra-atomic bremsstrahlung (IAB), investigating scintillation response of various materials and optimizing PMT and shielding configurations for visible-light photon detection
- Performed data analysis and calculations to diagnose and reduce systematic noise, increasing our data collection rate by over a factor of 20
- Ran and supported experiments at CHESS and ESRF synchrotron facilities, collaborating directly with beamline scientists to take data efficiently during our beam-time windows

Project Support

Summer 2023

The Nova Group

Framingham, MA

- Supported several projects with data processing, concisely organizing information from over 25 hours of interviews with medical professionals and sorting through thousands of manufacturing companies using partnership criteria

ENGINEERING PROJECTS

Mechanical Subteam Lead, Summer Lead | CUAUV

Oct. 2023 - Oct. 2025

- Designed the main pressure vessel ("UHPV") for our 2025 vehicle, incorporating multiple nitrile O-ring seals of varying sizes across its housings, and produced manufacturing drawings for precision outsourced fabrication, optimizing for size, strength, and cable management
- Coordinated the full team during the summer of 2024, creating and monitoring timelines for critical projects, troubleshooting technical and logistical issues as they arise, and serving as the primary point of contact for the team during remote lab testing and the RoboSub 2024 competition (3rd place overall)
- Designed a downwards-facing camera enclosure rated to 10m depth (FoS 30) for our 2024 vehicle, machining nitrile O-ring seal interfaces to tolerances as tight as $\pm 0.002"$ and providing critical visual feedback for our manipulator and torpedo systems

SKILLS AND CLASSES

Design and Fabrication: Fusion 360 (Modeling, Assemblies, CAM), Solidworks, Ansys, FDM Printing (Bambu, Prusa), Laser Cutting, Machining (Manual and CNC), Soldering, GD&T

Software and Data: Python (NumPy, Matplotlib), MATLAB, Arduino (C++), Git/GitHub, Photodiodes and Photomultiplier Tubes, High-Voltage Amplifiers and Circuits, Pulse-Height Analyzers, Serial Communication (UART, I2C), Turbidity and Pressure Sensors

Classes: Dynamics, Fluid Mechanics, Thermodynamics, Statics and Mechanics of Materials, FEA/Computational

Methods, Aeronautics, Classical Mechanics, Circuits and Electronics, Graduate Experimental Physics, Electromagnetism

Other: Eagle Scout, Cornell Society of Physics Students, NYLT Leadership Academy Staff Volunteer (2019-2024)